



Modelling Financial Markets

Seminar 5: Conditional Volatility — GARCH Models



Daniel Traian PELE

Drd. Antoaneta Amza, antoaneta.amza@csie.ase.ro

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

Acronyms used in this material (1/2)

- ACF: AutoCorrelation Function
- AI: Artificial Intelligence
- AIC: Akaike Information Criterion
- AR: AutoRegressive (model)
- ARCH: AutoRegressive Conditional Heteroskedasticity
- ARMA: AutoRegressive Moving Average (model)
- BET: Bucharest Exchange Trading (index)
- BIC: Bayesian Information Criterion
- DM: Diebold–Mariano (test)
- EGARCH: Exponential GARCH
- EWMA: Exponentially Weighted Moving Average
- GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- GARCH-N: GARCH with Normal innovations
- GED: Generalised Error Distribution
- GJR: Glosten–Jagannathan–Runkle (asymmetric GARCH)
- GJR-GARCH: Glosten–Jagannathan–Runkle GARCH
- GW: Giacomini–White (test)
- HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- IGARCH: Integrated GARCH
- LM: Lagrange Multiplier
- LR: Likelihood Ratio (test)
- MCS: Model Confidence Set
- MLE: Maximum Likelihood Estimation
- MZ: Mincer–Zarnowitz (regression)
- PIT: Probability Integral Transform
- QLIKE: Quasi-LIKElihood (loss function)

Acronyms used in this material (2/2)

- QMLE: Quasi-Maximum Likelihood Estimator
- US: United States
- VIX: CBOE Volatility Index

Outline

Introduction

Part A: Derivations

Part B: Computation and Inference

Part C: Open Question

Wrap-up

References

Seminar Objectives

By the end of this seminar you can

- Run the GARCH variance recursion and its forecasts step by step
- Estimate GARCH, GJR-GARCH and heavy-tailed variants and diagnose the residuals
- Compare models with likelihood ratios, information criteria and out-of-sample losses
- Measure estimation uncertainty with the delta method and a parametric bootstrap, and test on the boundary

- Background: Lecture 5; textbook [Franke, Härdle & Hafner \(2019\)](#)

Structure

- **Part A:** six derivations (A1–A6): multi-step variance, GJR persistence, delta method, moments, robust vs classical QMLE variance, ACF of squares
- **Part B:** eight computational exercises with inference (B1–B8)
- **Part C:** one open question, a seed for the team project (C1), a critique of an AI answer (C2) and a project seed from the lecture case study (C3)

How to Work in This Seminar

- **[Solved]** problems show the full worked solution; **[Proposed]** problems show only the task: their solutions are discussed in class
- Every proposed problem names its **model**: the solved problem that uses the same technique
 - ▶ Every Part B exercise ends with interpretation questions: ask what would change the conclusion
- Report every estimate with its uncertainty: robust standard error, bootstrap interval or p -value
- Check every fitted model: convergence, parameters at a bound, standardised residuals

Solved and Proposed Problems

Solved: worked examples in this deck

- Part A: A1 (multi-step variance: derivation and numbers), A3 (long-run variance, half-life, delta method)
- Part B: B1 (GARCH-t with diagnostics), B2 (GJR vs GARCH, LR test), B5 (QLIKE, DM, MZ, Giacomini–White, MCS), B7 (parametric bootstrap, IGARCH test under H_0)

Proposed: you solve them

- Part A: A2 (model: A1), A4 (model: A3), A5 (model: A3), A6 (model: A1)
- Part B: B3 (model: B2), B4 (model: B1, B2), B6 (model: B5), B8 (model: B7)
- Part C: C1, open question and project seed; C2, critique of an AI answer (model: A3); C3, quasi score-driven GARCH on 11 US stocks (lecture case study; model: B1, B2)

Setup: Python Environment and Data

-  **Open in Colab:** `chapter5_seminar_notebook.ipynb`
- Google Colab (recommended) or local Python ≥ 3.10 with `numpy`, `pandas`, `scipy`, `statsmodels`, `matplotlib` and `arch`
- Series: S&P 500, BET, Bitcoin; daily closes up to 18 September 2026
- Returns in %: $r_t = 100 (\ln P_t - \ln P_{t-1})$; equity indices on weekdays, Bitcoin 7 days a week
- Part C joins the BET and S&P 500 prices on common days first, then computes returns

Formulas Used Today

- GARCH(1,1) Bollerslev (1986): $\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$; GJR Glosten, Jagannathan & Runkle (1993): add $\gamma I(\varepsilon_{t-1} < 0) \varepsilon_{t-1}^2$
- Long-run variance $\sigma^2 = \omega / (1 - \alpha - \beta)$; forecast $\mathbb{E}_t \sigma_{t+h}^2 = \sigma^2 + (\alpha + \beta)^{h-1} (\sigma_{t+1}^2 - \sigma^2)$
- Half-life $h_{1/2} = \ln 0.5 / \ln(\alpha + \beta)$; $LR = 2(\ell_1 - \ell_0) \sim \chi_m^2$, or $\frac{1}{2}\chi_0^2 + \frac{1}{2}\chi_1^2$ for one parameter on the boundary (regular likelihood, identified nuisance parameters)
- Delta method: $\widehat{\text{s.e.}} g(\hat{\theta}) = \sqrt{\nabla g' \hat{V} \nabla g}$; Gaussian QMLE of (ω, α, β) , known mean: $\text{avar} = (\kappa_z - 1)J^{-1}$
- QLIKE loss $r_t^2/h_t + \ln h_t$ Patton (2011); Diebold–Mariano t on the loss differential Diebold & Mariano (1995)
- Sign-bias regression $\hat{z}_t^2$ on S_{t-1}^- , $S_{t-1}^- \hat{\varepsilon}_{t-1}$, $S_{t-1}^+ \hat{\varepsilon}_{t-1}$ Engle & Ng (1993)

A1: Multi-Step Variance Forecast (1/2) [Solved]

Context and data

- GARCH(1,1): $\varepsilon_t = \sigma_t z_t$, $\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$, with z_t i.i.d., mean 0 and variance 1, independent of the past
- Persistence $p = \alpha + \beta < 1$, long-run variance $\bar{\sigma}^2 = \omega / (1 - p)$; $\mathbb{E}_t$ is the expectation given the information at the close of day t
- Numbers (returns in %): $\omega = 0.05$, $\alpha = 0.10$, $\beta = 0.85$, start $\sigma_1^2 = \bar{\sigma}^2$; shocks $\varepsilon_1 = -2.0$, $\varepsilon_2 = 0.5$, $\varepsilon_3 = 3.0$, $\varepsilon_4 = -1.0$

What to do

- (a) Take conditional expectations of the recursion and derive $\mathbb{E}_t \sigma_{t+h}^2 = \bar{\sigma}^2 + p^{h-1}(\sigma_{t+1}^2 - \bar{\sigma}^2)$ for $h \geq 1$.
- (b) Sum the forecasts and derive the closed form of the H -day variance $V_H = \sum_{h=1}^H \mathbb{E}_t \sigma_{t+h}^2$.
- (c) Derive the sign of $V_H - H\sigma_{t+1}^2$, the error of the square-root-of-time rule, as a function of $\sigma_{t+1}^2 - \bar{\sigma}^2$.
- (d) Compute $\sigma_2^2, \dots, \sigma_5^2$ step by step and the forecasts $\mathbb{E}_4 \sigma_{4+h}^2$ for $h = 1, \dots, 5$.

A1: Multi-Step Variance Forecast (2/2) [Solved]

What to do

- (e) Compute V_5 from the sum and from the formula of (b); compare it with $5\sigma_5^2$. (f) Compute the half-life $\ln 0.5 / \ln p$.

What to report and interpret

- Report the two formulas, the four variances, the five forecasts, V_5 with the 5-day volatility $\sqrt{V_5}$ and the half-life.
- Interpretation: after the shock of +3%, does the square-root-of-time rule overstate or understate the 5-day risk, and why?

A1: Multi-Step Variance Forecast: Solution (1/2) [Solved]

- (a) Step 1: shift the recursion, $\sigma_{t+h}^2 = \omega + \alpha \varepsilon_{t+h-1}^2 + \beta \sigma_{t+h-1}^2$, and take $\mathbb{E}_t$.
- Step 2: for $h \geq 2$, $\mathbb{E}_t \varepsilon_{t+h-1}^2 = \mathbb{E}_t[\sigma_{t+h-1}^2 \mathbb{E}(z_{t+h-1}^2)] = \mathbb{E}_t \sigma_{t+h-1}^2$, hence $\mathbb{E}_t \sigma_{t+h}^2 = \omega + p \mathbb{E}_t \sigma_{t+h-1}^2$.
- Step 3: subtract $\bar{\sigma}^2 = \omega + p\bar{\sigma}^2$; the gap to the long run shrinks by p per step, and iterating from the known σ_{t+1}^2 gives the formula.
- (b) Summing, $V_H = H\bar{\sigma}^2 + (\sigma_{t+1}^2 - \bar{\sigma}^2) \sum_{j=0}^{H-1} p^j = H\bar{\sigma}^2 + (\sigma_{t+1}^2 - \bar{\sigma}^2) \frac{1-p^H}{1-p}$.
- (c) Subtracting $H\sigma_{t+1}^2$: $V_H - H\sigma_{t+1}^2 = (\bar{\sigma}^2 - \sigma_{t+1}^2) \left[H - \frac{1-p^H}{1-p} \right]$, and the bracket equals $\sum_{j=0}^{H-1} (1-p^j) > 0$ for $H \geq 2$.
- Interpretation: the rule $H\sigma_{t+1}^2$ understates the H -day variance when today is calmer than the long run and overstates it after turbulent days, because the variance reverts to $\bar{\sigma}^2$.

A1: Multi-Step Variance Forecast: Solution (2/2) [Solved]

- (d) Start: $\sigma_1^2 = 0.05/(1 - 0.95) = 1.0000$; then
 $\sigma_2^2 = \omega + \alpha \varepsilon_1^2 + \beta \sigma_1^2 = 0.05 + 0.10 \times (-2.0)^2 + 0.85 \times 1.0000 = 0.05 + 0.4000 + 0.8500 = 1.3000$.
- In the same way $\sigma_3^2 = 0.05 + 0.0250 + 1.1050 = 1.1800$, $\sigma_4^2 = 0.05 + 0.9000 + 1.0030 = 1.9530$ and
 $\sigma_5^2 = 0.05 + 0.1000 + 1.6600 = 1.8100$.
- Forecasts: the gap $\sigma_5^2 - \bar{\sigma}^2 = 0.8101$ shrinks by 0.95 per day, so $\mathbb{E}_4 \sigma_{4+h}^2 = 1.8100, 1.7695, 1.7311, 1.6945, 1.6598$ for $h = 1, \dots, 5$.
- (e) Formula: $V_5 = 5 \times 1.0000 + 0.8101 \times (1 - 0.95^5)/(1 - 0.95) = 5 + 0.8101 \times 4.524 = 8.665$, the same as the sum;
 5-day volatility $\sqrt{8.665} = 2.944\%$.
- The square-root-of-time rule gives $5\sigma_5^2 = 9.050 > V_5$: after the turbulent days it overstates the 5-day variance, as (c) predicts for $\sigma_5^2 > \bar{\sigma}^2$.
- (f) Half-life = $\ln 0.5 / \ln 0.95 = -0.6931 / (-0.0513) = 13.5$ days: half of any variance shock is gone after about two weeks of trading.

A2: GJR Persistence with Skewed Innovations (1/2) [Proposed]

Solution discussed in class

Context and data

- GJR-GARCH(1,1) [Glosten, Jagannathan & Runkle \(1993\)](#): $\sigma_t^2 = \omega + (\alpha + \gamma I_{t-1})\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2$, $I_{t-1} = 1(\varepsilon_{t-1} < 0)$; z_t i.i.d., mean 0, variance 1, not necessarily symmetric
- Data for (c): S&P 500 (data/market/GSPC.INDX.csv), daily closes 2000–2026, 6 714 daily log returns in %, $r_t = 100(\ln P_t - \ln P_{t-1})$; GJR-GARCH(1,1) with skewed-t innovations [Hansen \(1994\)](#): $\hat{\alpha} = 0.00000$, $\hat{\gamma} = 0.21116$, $\hat{\beta} = 0.87965$, $\hat{\eta} = 7.70$, $\hat{\lambda} = -0.150$

What to do

- (a) Take expectations of the recursion and show that the persistence is $\alpha + \gamma m_- + \beta$, with $m_- = \mathbb{E}[z_t^2 1(z_t < 0)]$.
- (b) Show that $m_- = 1/2$ when z_t is symmetric, and explain why m_- differs in general from $P(z_t < 0)$.
- (c) Compute m_- and $P(z_t < 0)$ by numerical integration of the fitted skewed-t density; compute the persistence and the half-life with m_- and with $1/2$.

A2: GJR Persistence with Skewed Innovations (2/2) [Proposed]

Solution discussed in class

What to report and interpret

- Report m_- , $P(z_t < 0)$, both persistences and both half-lives. Interpretation: what goes wrong if the software formula $\alpha + \gamma/2 + \beta$ is used with skewed innovations?
- Model: A1

A3: Long-Run Volatility, Half-Life and Delta Method (1/2) [Solved]

Context and data

- S&P 500 (data/market/GSPC.INDX.csv), daily closes 2000–2026, 6 714 daily log returns in %, $r_t = 100 (\ln P_t - \ln P_{t-1})$
- GARCH(1,1) with Student-t innovations (Lecture 5): $\hat{\omega} = 0.015714$, $\hat{\alpha} = 0.122573$, $\hat{\beta} = 0.872139$; robust covariance of $(\hat{\omega}, \hat{\alpha}, \hat{\beta})$ from the fit, with $\widehat{\text{s.e.}}(\hat{\alpha} + \hat{\beta}) = 0.0047$; 251.4 trading days per year

What to do

- (a) Derive the long-run variance $\sigma^2 = \omega / (1 - \alpha - \beta)$ by taking unconditional expectations in the recursion.
- (b) Compute the long-run daily and annual volatility and the half-life; compare with the sample volatility 19.2% p.a.
- (c) Compute the gradient of $\sigma^2(\omega, \alpha, \beta)$ and, by the delta method, the standard error of the annual long-run volatility.
- (d) Compute the delta-method standard error of the half-life $h = \ln 0.5 / \ln p$, $p = \alpha + \beta$, and build a 95% interval for h that respects $h > 0$.

A3: Long-Run Volatility, Half-Life and Delta Method (2/2) [Solved]

What to report and interpret

- Report the long-run volatility with its standard error, the half-life with its standard error and the transformed interval.
- Interpretation: why is the delta method useless for the half-life, and what does that mean for the question $p = 1$ (B7)?

A3: Long-Run Volatility, Half-Life and Delta Method: Solution (1/2) [Solved]

- (a) Under covariance stationarity $\mathbb{E}\sigma_t^2 = \mathbb{E}\sigma_{t-1}^2 = \mathbb{E}\varepsilon_{t-1}^2 = \sigma^2$, so $\sigma^2 = \omega + (\alpha + \beta)\sigma^2$ and $\sigma^2 = \omega/(1 - \alpha - \beta)$.
- (b) $\hat{\rho} = \hat{\alpha} + \hat{\beta} = 0.994712$; $\hat{\sigma}^2 = 0.015714/(1 - 0.994712) = 2.972$ (%² per day).
- Daily volatility $\sqrt{2.972} = 1.724\%$; annual $1.724 \times \sqrt{251.4} = 27.3\%$, against the sample value 19.2%.
- Half-life $\ln 0.5 / \ln 0.994712 = -0.6931 / (-0.005302) = 130.7$ days.
- Interpretation: the long-run volatility exceeds the sample value because dividing by $1 - \hat{\rho} = 0.005288$ magnifies small errors in $\hat{\rho}$; the next step measures how much.

A3: Long-Run Volatility, Half-Life and Delta Method: Solution (2/2) [Solved]

- (c) Gradient $\nabla \sigma^2 = \left(\frac{1}{1-p}, \frac{\omega}{(1-p)^2}, \frac{\omega}{(1-p)^2} \right) = (189.1, 562.0, 562.0)$; $\widehat{\text{s.e.}}(\hat{\sigma}^2) = \sqrt{\nabla' \hat{V} \nabla} = 2.35$.
- Annual volatility $v = \sqrt{251.4 \sigma^2}$ has $\partial v / \partial \sigma^2 = 251.4 / (2v)$, so $\widehat{\text{s.e.}}(\hat{v}) = 10.8$ points around 27.3%: a 95% band of about ± 21 points.
- (d) $|\partial h / \partial p| = -\ln 0.5 / [p(\ln p)^2] = 24\,789$, so $\widehat{\text{s.e.}}(\hat{h}) = 24\,789 \times 0.0047 = 116$ days for $\hat{h} = 130.7$ days.
- A symmetric interval $\hat{h} \pm 1.96 \times 116$ would include negative half-lives: the linearisation fails because $h(p)$ explodes as $p \rightarrow 1$.
- Instead, h is increasing in p : map $\hat{p} \pm 1.96 \times 0.0047 = [0.9855; 1.0039]$ to $h \in [48; \infty)$, since the upper end passes $p = 1$.
- Interpretation: the data bound the half-life from below only; whether $p = 1$ is plausible needs a proper test on the boundary (B7).

A4: Finiteness of the Fourth Moment (1/2) [Proposed]

Solution discussed in class

Context and data

- S&P 500 (data/market/GSPC.INDX.csv), daily closes 2000–2026, 6 714 daily log returns in %, $r_t = 100 (\ln P_t - \ln P_{t-1})$
- GARCH(1,1) with Normal innovations (GARCH-N): $\hat{\alpha} = 0.120604$, $\hat{\beta} = 0.859878$; sample kurtosis of the returns 13.7
- GARCH(1,1)-t: $\hat{\alpha} = 0.1226$, $\hat{\beta} = 0.8721$, $\hat{\nu} = 6.256$; with Normal z_t , $K = 3[1 - (\alpha + \beta)^2]/[1 - (\alpha + \beta)^2 - 2\alpha^2]$ if the denominator is positive (Seminar 1, A9)

What to do

- (a) Check the condition $(\alpha + \beta)^2 + 2\alpha^2 < 1$ for GARCH-N and compute the implied kurtosis K .
- (b) Repeat (a) with $\hat{\alpha}$ and $\hat{\beta}$ rounded to 3 decimals and compare both values of K with the sample kurtosis.
- (c) For Student-t innovations, compute $\kappa_z = 3(\nu - 2)/(\nu - 4)$ and check the condition $(\alpha + \beta)^2 + (\kappa_z - 1)\alpha^2 < 1$ for the GARCH-t estimates.

A4: Finiteness of the Fourth Moment (2/2) [Proposed]

Solution discussed in class

What to report and interpret

- Report both conditions, the two values of K and κ_z . Interpretation: can the kurtosis of the returns be used to validate a fitted GARCH-t?
- Model: A3

A5: Robust versus Classical Variance of the Gaussian QMLE, ARCH(1) (1/2) [Proposed]

Solution discussed in class

Context and data

- ▣ ARCH(1): $\varepsilon_t = \sigma_t z_t$, $\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2$ with ω known; z_t i.i.d., mean 0, variance 1, kurtosis $\kappa_z = \mathbb{E}z_t^4 < \infty$, not necessarily Normal
- ▣ Gaussian quasi-log-likelihood per day $\ell_t(\alpha) = -\frac{1}{2}(\ln \sigma_t^2 + \varepsilon_t^2/\sigma_t^2)$; notation $x_t = \varepsilon_{t-1}^2/\sigma_t^2$. Data for (d): the S&P 500 GARCH-N residual kurtosis 4.73 (Lecture 5)

What to do

- ▣ (a) Derive the score $s_t = \partial \ell_t / \partial \alpha$ and its variance $B = \mathbb{E}s_t^2$.
- ▣ (b) Derive the second derivative $\partial^2 \ell_t / \partial \alpha^2$ and $A = -\mathbb{E} \partial^2 \ell_t / \partial \alpha^2$.
- ▣ (c) Show that the sandwich variance $A^{-1}BA^{-1}$ equals $\frac{\kappa_z - 1}{2}A^{-1}$, where A^{-1} is the classical (inverse-Hessian) variance.
- ▣ (d) Compute the ratio of robust to classical standard errors for Student-t innovations with $\nu = 6.26$ and for $\hat{\kappa}_z = 4.73$.

A5: Robust versus Classical Variance of the Gaussian QMLE, ARCH(1) (2/2) [Proposed]

Solution discussed in class

What to report and interpret

- Report s_t , A , B , the variance ratio and the two numerical ratios; compare with the observed ratio of 1.37 in Lecture 5.
- Interpretation: what happens to this ratio, and to the $\sqrt{n}$ theory, when $\kappa_z = \infty$? Model: A3

A6: Autocorrelation of Squared Shocks (1/2) [Proposed]

Solution discussed in class

Context and data

- ▣ GARCH(1,1) with $\mathbb{E}\varepsilon_t^4 < \infty$ is an ARMA(1,1) in ε_t^2 : $\varepsilon_t^2 = \omega + (\alpha + \beta)\varepsilon_{t-1}^2 + \nu_t - \beta\nu_{t-1}$, with $\nu_t = \varepsilon_t^2 - \sigma_t^2$ (Lecture 5)
- ▣ For an ARMA(1,1) $y_t = \phi y_{t-1} + u_t + \theta u_{t-1}$: $\rho_1 = (1 + \phi\theta)(\phi + \theta)/(1 + 2\phi\theta + \theta^2)$ and $\rho_k = \phi^{k-1}\rho_1$
- ▣ Data for (b): S&P 500 GARCH-N, $\hat{\alpha} = 0.1206$, $\hat{\beta} = 0.8599$; sample autocorrelations of the squared demeaned returns 0.313 (lag 1) and 0.248 (lag 10)

What to do

- ▣ (a) Substitute $\phi = \alpha + \beta$ and $\theta = -\beta$ and derive $\rho_1(\varepsilon^2) = \alpha(1 - \beta^2 - \alpha\beta)/(1 - \beta^2 - 2\alpha\beta)$ and $\rho_k = \rho_1(\alpha + \beta)^{k-1}$.
- ▣ (b) Evaluate ρ_1 and ρ_{10} for the S&P 500 estimates and compare them with the sample values.
- ▣ (c) Explain why the gap can be large even if the model is correct, using the moment condition of A4.

A6: Autocorrelation of Squared Shocks (2/2) [Proposed]

Solution discussed in class

What to report and interpret

- Report the formula, ρ_1 , ρ_{10} and the sample values. Interpretation: is matching the autocorrelation function of r_t^2 a strong or a weak check of a GARCH fit?
- Model: A1

B1: GARCH-t on the S&P 500 with Diagnostics (1/2) [Solved]

Context and data

- S&P 500 (data/market/GSPC.INDX.csv), daily closes 2000–2026, 6 714 daily log returns in %, $r_t = 100 (\ln P_t - \ln P_{t-1})$
- Mean equation $r_t = \mu + \varepsilon_t$; variance equation GARCH(1,1) [Bollerslev \(1986\)](#); innovations Student-t with ν degrees of freedom, scaled to unit variance

What to do

- (a) Test for ARCH effects before modelling: Ljung–Box $Q(10)$ on r_t^2 and Engle's LM test with 5 lags on the demeaned returns, at the 5% level.
- (b) Estimate GARCH(1,1)-t by maximum likelihood (`arch_model(r, p=1, q=1, dist='t')`); report robust [Bollerslev & Wooldridge \(1992\)](#) and classical standard errors and their ratio.
- (c) Diagnose $\hat{z}_t = \hat{\varepsilon}_t / \hat{\sigma}_t$ at 5%: $Q(10)$ on $\hat{z}_t$ and $\hat{z}_t^2$, LM(5), and the three sign-bias t -tests plus their joint χ^2_3 test [Engle & Ng \(1993\)](#).

B1: GARCH-t on the S&P 500 with Diagnostics (2/2) [Solved]

What to report and interpret

- Report a table of estimates with both standard errors and a table of test statistics with p -values and decisions.
- Interpretation: which diagnostics pass and which fail, what does the failure suggest, and why do robust and classical standard errors differ here?

B1: GARCH-t on the S&P 500 with Diagnostics: Solution (1/2) [Solved]

	$\hat{\omega}$	$\hat{\alpha}$	$\hat{\beta}$	$\hat{\nu}$	$\hat{\alpha} + \hat{\beta}$
Estimate	0.0157	0.1226	0.8721	6.26	0.9947
Robust s.e.		0.0107	0.0103	0.48	
Classical s.e.		0.0101	0.0096		

- (a) Before modelling, $Q(10) = 6133$ and $LM(5) = 1684$ against 5% critical values 18.3 (χ^2_{10}) and 11.1 (χ^2_5); both $p < 0.001$, so no-ARCH is rejected.
- (b) Reading the table: $\hat{\beta} = 0.8721$ carries yesterday's variance, $\hat{\alpha} = 0.1226$ adds the news, $\hat{\nu} = 6.26$ signals heavy tails; persistence 0.9947.
- The robust/classical ratio is 1.06 for $\hat{\alpha}$ and 1.08 for $\hat{\beta}$: the Student-t likelihood is itself misspecified (skewness -0.58); here the sandwich is larger. It covers uncertainty around the pseudo-true parameter only: with a misspecified non-Gaussian density the QMLE of the variance scale need not be consistent [Fan, Qi & Xiu \(2014\)](#), so the Student-t estimates are read structurally only if the density is right.

B1: GARCH-t on the S&P 500 with Diagnostics: Solution (2/2) [Solved]

- (c) Variance equation: $Q(10)$ on $\hat{z}_t^2 = 13.6$ ($p = 0.192$) and $LM(5) = 5.8$ ($p = 0.326$): not rejected at 5%: no remaining clustering is detected (naive χ^2 laws, see the caveat below).
- Mean equation: $Q(10)$ on $\hat{z}_t = 21.9$ ($p = 0.016$): rejected at 5%; a small autocorrelation remains that a constant mean ignores.
- Asymmetry: sign bias $t = 3.56$, negative size bias $t = 1.13$, positive size bias $t = -2.61$; joint $\chi^2_3 = 40.7$ against 7.81: rejected, asymmetry is left in the residuals.
- Caveat: with estimated parameters these χ^2 laws are only approximate [Li & Mak \(1994\)](#); here the rejections are far from the critical values, so the decisions stand.
- Interpretation: the variance dynamics are captured, but a symmetric GARCH misses the leverage effect; the next step is GJR-GARCH (B2).

B2: GJR vs GARCH on the S&P 500 (1/2) [Solved]

Context and data

- S&P 500 (data/market/GSPC.INDX.csv), daily closes 2000–2026, 6 714 daily log returns in %, $r_t = 100 (\ln P_t - \ln P_{t-1})$
- GJR-GARCH(1,1)-t [Glosten, Jagannathan & Runkle \(1993\)](#): $\sigma_t^2 = \omega + (\alpha + \gamma I_{t-1})\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2$, $I_{t-1} = 1(\varepsilon_{t-1} < 0)$; GARCH(1,1)-t of B1 is the special case $\gamma = 0$

What to do

- (a) Estimate GJR-GARCH(1,1)-t and report $\hat{\alpha}$, $\hat{\gamma}$ with its robust t -statistic, and $\hat{\beta}$.
- (b) Test $H_0: \gamma = 0$ with $LR = 2(\ell_{GJR} - \ell_{GARCH})$ against χ_1^2 at 5%.
- (c) Compare AIC and BIC of the two models and repeat the joint sign-bias test of B1 on the GJR residuals.
- (d) Measure the economic size of the asymmetry with the news impact curve: next-day variance after shocks of -3% and $+3\%$.

What to report and interpret

- Report the estimates, the LR test (statistic, p -value, decision), ΔAIC , ΔBIC and the two news-impact values.

B2: GJR vs GARCH on the S&P 500 (2/2) [Solved]

What to report and interpret

- Interpretation: why is $\hat{\alpha}$ exactly 0, and is the χ^2_1 law of the LR test still valid?

B2: GJR vs GARCH on the S&P 500: Solution [Solved]

- (a) $\hat{\alpha} = 0.000$, $\hat{\gamma} = 0.205$ (robust $t = 10.03$), $\hat{\beta} = 0.881$: good news has slope $\hat{\alpha}$, bad news $\hat{\alpha} + \hat{\gamma}$.
- (b) $LR = 2(-8944.647 - (-9062.473)) = 235.65$, $p < 0.001$, far above 3.84: H_0 is rejected at 5%; the χ^2_1 law assumes a correct Student-t density (B1 finds skewness), and the robust t of (a), which gives the same verdict, is $N(0, 1)$ only if the other parameters are interior (here $\hat{\alpha} = 0$).
- (c) $\Delta AIC = -233.7$ and $\Delta BIC = -226.8$ (GJR minus GARCH): both prefer GJR; the joint sign-bias χ^2_3 falls from 40.7 to 28.1, still above 7.81.
- (d) From the sample variance 1.47, a -3% shock gives next-day variance 3.16 and a $+3\%$ shock 1.31: the forecast after bad news is 2.4 times that after good news, and the $+3\%$ shock even lowers the variance ($\hat{\alpha} = 0$).
- $\hat{\alpha} = 0$ sits on the bound $\alpha \geq 0$: positive shocks do not raise volatility; $\gamma = 0$ is interior, so χ^2_1 holds if the true $\alpha > 0$, and if $\alpha_0 = 0$ the law is non-standard [Andrews \(2001\)](#), yet $LR = 235.65$ rejects under any reasonable law.

B3: GJR vs GARCH on the BET [Proposed]

Solution discussed in class

Context and data

- BET (data/market/BET.csv), daily closes 2000–2026, 6 670 daily log returns in %
- The same two models as in B2: GARCH(1,1)-t and GJR-GARCH(1,1)-t, constant mean, Student-t innovations

What to do

- (a) Estimate both models and report $\hat{\gamma}$ with its robust t -statistic.
- (b) Test $H_0: \gamma = 0$ by LR against χ^2_1 at the 5% and 1% levels.
- (c) Compute ΔAIC and ΔBIC (GJR minus GARCH) and the BIC penalty per parameter, $\ln n$.

What to report and interpret

- Report the estimates, the LR decision at both levels and the two criteria. Interpretation: do LR and BIC agree, why can they disagree, and is the leverage effect weaker on the BET than on the S&P 500?
- Model: B2

B4: Innovation Distribution for Bitcoin (1/2) [Proposed]

Solution discussed in class

Context and data

- Bitcoin (data/market/BTC-USD.CC.csv), daily closes 2014–2026, 4 384 daily log returns in %, 365 days a year
- GARCH(1,1) with constant mean; four innovation laws: Normal, Student-t, skewed-t [Hansen \(1994\)](#)(asymmetry λ) and GED

What to do

- (a) Estimate the four models and compare their log-likelihoods and BIC.
- (b) Test $H_0: \lambda = 0$ (skewed-t against Student-t) by LR against χ^2_1 at 5%.
- (c) Report the applicable shape and skewness parameters with their standard errors, and $\hat{\alpha} + \hat{\beta}$, for each law; the Normal distribution has none, and the GED shape is not a degrees-of-freedom parameter; state which parameters sit at a bound.
- (d) Test the Normal distribution ($1/\nu = 0$) against the Student-t distribution by LR with the boundary law $\frac{1}{2}\chi^2_0 + \frac{1}{2}\chi^2_1$ (5% critical value 2.71).

B4: Innovation Distribution for Bitcoin (2/2) [Proposed]

Solution discussed in class

What to report and interpret

- Report a table of ℓ , BIC and persistence, and the two LR tests. Interpretation: the sample skewness of $\hat{z}_t$ is negative, so why does the likelihood find no skewness? With $\hat{\nu} < 4$, is Gaussian-QMLE inference valid for Bitcoin [Hall & Yao \(2003\)](#), and how does it differ from inference for the Student-t MLE of B1?
- Model: B1, B2

B5: Forecast Comparison on the S&P 500 (1/2) [Solved]

Context and data

- S&P 500 (data/market/GSPC.INDX.csv), daily closes 2000–2026, 6 714 daily log returns in %, $r_t = 100 (\ln P_t - \ln P_{t-1})$; forecast period 2016–2026 (2 692 one-day-ahead forecasts)
- Models: GARCH-N, GARCH-t, GJR-t, EGARCH-t re-estimated every January on all data since 2000 (expanding window); EWMA with $\lambda = 0.94$ J.P. Morgan/Reuters (1996); Hist-20, the mean of the last 20 squared returns
- Proxy of the true variance: r_t^2 , ignoring the mean ($\hat{\mu}^2 = 0.0062$ against an average variance of 1.47); loss QLIKE $= r_t^2 / h_t + \ln h_t$ Patton (2011), with h_t the variance forecast

What to do

- (a) Compute the mean QLIKE of each model.
- (b) Test equal accuracy against GARCH-t with the Diebold–Mariano t -statistic on $d_t = L_t^m - L_t^{GARCH-t}$, HAC standard error with 5 lags, at 5% Diebold & Mariano (1995).
- (c) Run the Mincer–Zarnowitz regression $r_t^2 = a + b h_t + u_t$ Mincer & Zarnowitz (1969) with HAC errors and the Wald test of $a = 0$, $b = 1$.

B5: Forecast Comparison on the S&P 500 (2/2) [Solved]

What to do

- (d) Repeat the comparison with a rolling 1 000-day window, re-estimated each January, and the Giacomini–White test [Giacomini & White \(2006\)](#) (instruments 1, d_t).
- (e) Compute the 90% Model Confidence Set [Hansen, Lunde & Nason \(2011\)](#) on the QLIKE losses of the expanding-window forecasts of (a): statistic T_R , stationary bootstrap with mean block length 10, 10 000 draws, fixed seed.

What to report and interpret

- Report two tables: QLIKE, DM t , MZ b , R^2 and Wald p ; then GW statistic, p -value and MCS p -value.
- Interpretation: which models are significantly worse than GARCH-t, why is R^2 low, why is DM not valid for GJR-t vs GARCH-t, and do DM and MCS agree?

B5: Forecast Comparison on the S&P 500: Solution (1/2) [Solved]

Model	QLIKE	DM t	MZ b	MZ R^2	Wald p
GARCH-N	0.7510	-1.47	0.97	0.250	0.995
GARCH-t	0.7570	—	0.88	0.238	0.804
GJR-t	0.7319	-1.22	0.84	0.276	0.680
EGARCH-t	0.7454	-0.36	1.24	0.293	0.713
EWMA	0.8149	3.48	0.89	0.172	0.685
Hist-20	0.8888	4.10	0.68	0.152	0.028

- Procedure: forecast h_t each day with parameters from the last January, compute L_t , average over 2 692 days, and regress d_t on a constant with HAC errors.
- (a–b) Lowest QLIKE: GJR-t; only EWMA ($t = 3.48$) and Hist-20 ($t = 4.10$) exceed 1.96: they are significantly worse at 5%.
- (c) Unbiasedness is not rejected for the GARCH models (Wald $p > 0.6$) but is rejected for Hist-20 ($p = 0.028$); R^2 is low because $r_t^2 \approx \sigma_t^2 z_t^2$ is a noisy proxy.

B5: Forecast Comparison on the S&P 500: Solution (2/2) [Solved]

Model	t (rolling)	GW	p	MCS p
GARCH-N	1.87	5.85	0.054	0.586
GARCH-t	—	—	—	0.353
GJR-t	−0.18	2.49	0.288	1.000
EGARCH-t	0.07	3.21	0.201	0.586
EWMA	3.20	11.40	0.003	0.008
Hist-20	3.87	16.41	< 0.001	0.001

- (d) Why GW: with nested models, estimated parameters and an expanding window the DM differential degenerates under H_0 , so DM is not $N(0, 1)$; a rolling window keeps estimation noise and makes GW valid.
- $GW = n\bar{Z}'\hat{\Omega}^{-1}\bar{Z} \sim \chi_2^2$ (critical value 5.99): rejected for EWMA and Hist-20 only; GARCH-N is borderline ($p = 0.054$).
- (e) MCS on the expanding-window forecasts: a model stays in the 90% set if its MCS p -value exceeds 0.10; the set is GARCH-N, GARCH-t, GJR-t, EGARCH-t, the same verdict as DM, now with the error rate controlled over six models.

B6: Forecast Comparison on the BET (1/2) [Proposed]

Solution discussed in class

Context and data

- BET (data/market/BET.csv), daily closes 2000–2026, 6 670 daily log returns in %; forecast period 2016–2026 (2 675 one-day-ahead forecasts)
- The same six models, re-estimation scheme, proxy r_t^2 and QLIKE loss as in B5

What to do

- (a) Compute the mean QLIKE of each model and the Diebold–Mariano t -statistic against GARCH-t (HAC, 5 lags), at 5%.
- (b) Run the Mincer–Zarnowitz regression and the Wald test of $a = 0$, $b = 1$ for each model.
- (c) Compute the Giacomini–White test with a rolling 1 000-day window and, as in B5 (e), the 90% Model Confidence Set on the expanding-window QLIKE losses.

What to report and interpret

- Report the two tables of B5 for the BET. Interpretation: does the ranking change between the S&P 500 and the BET, do DM and MCS give the same verdict for EWMA, and which part of the BET model would you improve first?

B6: Forecast Comparison on the BET (2/2) [Proposed]

Solution discussed in class

What to report and interpret

- Model: B5

B7: How Precise Is the Half-Life? Bootstrap and IGARCH Test (1/2) [Solved]

Context and data

- S&P 500 (data/market/GSPC.INDX.csv), daily closes 2000–2026, 6 714 daily log returns in %, $r_t = 100 (\ln P_t - \ln P_{t-1})$
- GARCH(1,1)-t of B1: $\hat{p} = \hat{\alpha} + \hat{\beta} = 0.9947$, half-life 131 days; estimation imposes $\alpha + \beta \leq 1$; random seed fixed for reproducibility

What to do

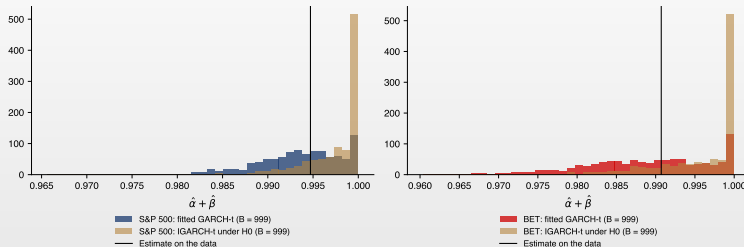
- (a) Simulate $B = 999$ paths of 6 714 days (500 burn-in days) from the fitted GARCH(1,1)-t and re-estimate the model on each path.
- (b) Compute the 95% percentile intervals of $\hat{p}^*$ and of the half-life, the bootstrap standard deviation of $\hat{p}^*$ and the share of draws at the bound $p = 1$; compare with the robust standard error.
- (c) Test $H_0: \alpha + \beta = 1$: fit IGARCH(1,1)-t, compute $LR = 2(\ell_{GARCH} - \ell_{IGARCH})$, simulate $B = 999$ paths **under** H_0 , re-estimate both models on each, and compute the bootstrap p -value $(1 + \#\{LR^* \geq LR\})/(B + 1)$.
- (d) Compare this p -value with the naive χ_1^2 and the boundary mixture $\frac{1}{2}\chi_0^2 + \frac{1}{2}\chi_1^2$; decide at 5% and 10%.

B7: How Precise Is the Half-Life? Bootstrap and IGARCH Test (2/2) [Solved]

What to report and interpret

- Report the histogram of $\hat{p}^*$ under both schemes, the intervals, the three p -values and the decision.
- Interpretation: why is the half-life interval so asymmetric, and why is “the percentile interval reaches 1” not a test of IGARCH?

B7: How Precise Is the Half-Life? Bootstrap and IGARCH Test: Solution (1/2) [Solved]



- (a–b) Blue: $\hat{\rho}^*$ from the fitted model; 95% interval [0.9833; 1.0000]; bootstrap s.d. 0.0047 against the robust s.e. 0.0047: close.
- Half-life from 41 days to ∞ : $\ln 0.5 / \ln x$ explodes as $x \rightarrow 1$ (delta-method s.e. 116 days, A3); 8.9% of draws sit at the bound, so the interval is truncated at 1 [Andrews \(2000\)](#).

B7: How Precise Is the Half-Life? Bootstrap and IGARCH Test: Solution (2/2) [Solved]

- (c) IGARCH-t fit: $\hat{\alpha}_0 = 0.127$, $\hat{\nu}_0 = 6.03$; $LR = 1.11$.
- Under H_0 (amber histogram), 52% of the re-estimates sit at $p = 1$, close to the half predicted by the mixture law; the simulated 95% critical value of LR is 3.36.
- (d) p -values: 0.29 (naive χ^2_1), 0.15 (mixture), 0.17 (bootstrap LR) and 0.17 (bootstrap $\hat{p}$).
- Decision: H_0 is not rejected at 5% or at 10%; the naive χ^2_1 roughly doubles the p -value, the mixture and the bootstrap agree.
- Economic conclusion: 27 years of data cannot tell a half-life of about 130 days from an infinite one; simulating from $\hat{\theta}$ describes precision, simulating from the null model is what a test needs.

B8: Bootstrap and IGARCH Test for the BET [Proposed]

Solution discussed in class

Context and data

- BET (data/market/BET.csv), daily closes 2000–2026, 6 670 daily log returns in %
- GARCH(1,1)-t with $\hat{p} = 0.9907$ (half-life 74 days); the same procedure and number of draws as in B7

What to do

- (a) Run the parametric bootstrap with $B = 999$ from the fitted model; report the 95% intervals for p and for the half-life and the share of draws at $p = 1$.
- (b) Compare the bootstrap standard deviation of $\hat{p}^*$ with the robust standard error of $\hat{\alpha} + \hat{\beta}$.
- (c) Test IGARCH by simulation under H_0 ($B = 999$) and compare with the mixture p -value, at 5%.

What to report and interpret

- Report the intervals, the two standard errors and the test. Interpretation: is the BET half-life shorter than the S&P 500 one with confidence, and what would a formal test of the difference need?
- Model: B7

C1: Is the Romanian Market's Volatility Driven by the US? [Proposed]

Solution discussed in class

The question

- Wall Street closes after the Bucharest Stock Exchange: yesterday's S&P 500 shock is news for today's BET
- Does u_{t-1}^2 , the squared lagged S&P 500 return, raise the conditional variance of the BET?
- Model: $\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2 + \delta u_{t-1}^2$, with $r_t = \mu + \phi u_{t-1} + \varepsilon_t$

Deliverable

- A Quantlet folder in the team repository and one slide with the conclusion and its main limitation

The answer should contain

- Prices joined on common trading days first, then returns
- The likelihood written step by step (Student-t), estimated by numerical optimisation
- An LR test of $\delta = 0$; note that $\delta \geq 0$ puts the null on the boundary
- Stability: two subperiods (e.g. before and after 2013)
- The model-based share of the BET variance generated by the US term, $\sum_{j \geq 0} \beta^j \delta u_{t-1-j}^2 / \sigma_t^2$, and how it changes over time

C2: Critique an AI Answer [Proposed]

Solution discussed in class

The prompt sent to an artificial intelligence (AI) assistant

- “px holds daily S&P 500 closes. Fit a GARCH(1,1) with Student- t innovations using arch and give me the annualised long-run volatility and the half-life of a shock.”

The assistant's answer

```
import numpy as np
from arch import arch_model
r = np.log(px).diff().dropna()          # daily log returns
am = arch_model(r, vol='GARCH', p=1, q=1, dist='t')
res = am.fit(displ='off')
om = res.params['omega']
a, b = res.params['alpha[1]'], res.params['beta[1]']
lr_var = om / (1 - a - b)               # long-run daily variance
ann_vol = lr_var * 252                  # annualised volatility
half_life = np.log(0.5) / np.log(b)    # half-life in days
```

- “For the S&P 500 this gives $\text{ann_vol} = 0.075$ (7.5% a year) and a half-life of 5.1 days.”

C2: Critique an AI Answer, the Task [Proposed]

Solution discussed in class

- The code runs, but the answer contains **three** mistakes typical for GARCH work
 - ▶ Everything else in the answer is correct
- For each mistake
 - ▶ Name the line and say why it is wrong
 - ▶ Say how you would detect it: a warning, a number from A3, a slide of the lecture
 - ▶ Write the corrected line(s) and the corrected numbers
- Log each mistake as in `AI_ERRORS.md`
 - ▶ What was asked, what the AI answered, what was wrong, how you found out, the correct version
- Hint: compare with A3 and with the slide “Common Pitfalls”

C3: Quasi Score-Driven GARCH on US Stocks [Proposed]

Solution discussed in class

Task

- ▣ Blasques, Francq & Laurent (2023); data/market: the 11 case-study stocks and the VIX, 3 January 1995 – 28 February 2019. Model: B1, B2.
- ▣ 1. Join each stock (adjusted_close) with the VIX (close) on common days, from the stock's first date; keep stocks with at least 4000 returns; compute $r_t = 100 \Delta \ln P_t$ and $X_t = \text{VIX}_t^2 / 252$ (same day as r_t).
- ▣ 2. Write step by step the log-likelihood of eqs. (31)–(32) ($c = 1000$, $\zeta \in (-1, 1/2)$: $\zeta = 0$ interior, LR against χ_1^2); AR(1) mean given r_1 ; f_1 = residual variance; standardised Student-t, $1/\xi$ degrees of freedom; estimate GARCH(1,1)-t ($\zeta = 0$), Beta-t-GARCH ($\zeta = \xi$) and the unrestricted model.
- ▣ 3. Report the estimates with standard errors, the log-likelihoods and the LR tests of $\zeta = 0$ and $\xi = \zeta$ (Table 5), then the rejection frequencies (Table 4).
- ▣ 4. Compute the naive PIT-mean statistic $|\sum_t u_t - T/2| / \sqrt{T/12}$ for the three models and how often it exceeds 1.96.

Interpretation

- ▣ The naive PIT statistic ignores estimation error (corrected in Section 5.1) and tests only $\mathbb{E}u_t = 1/2$: which departures can it detect, and which not?
- ▣ Re-estimate on 3 January 1995 – 18 September 2026: does $\hat{\zeta}$ change, and why?
- ▣ Is $\hat{\zeta} < \hat{\xi}$ for most of your stocks, as in Fig. 2? What does it say about the downweighting in Beta-t-GARCH?
- ▣ Centene: $\hat{\xi} = 0.258$, $\hat{\zeta} = 0.048$ (Table 5). For the six banks, which restriction costs less log-likelihood, $\zeta = 0$ or $\zeta = \xi$? Distinguish relative fit from rejection.
- ▣ Give a binomial interval for a rejection frequency, assuming hypothetically i.i.d. rejections; why can it not establish agreement with the 89.5% of Table 4 for 11 selected survivors exposed to common shocks?
- ▣ Deliverable: a Quantlet folder and one slide with the table, the tests and the main limitation.

Common Pitfalls

- ▣ Fitting GARCH to returns in decimals: the optimiser may stop far from the maximum; use % or rescale
- ▣ Reading classical standard errors when the innovations are not Normal; use robust ones
- ▣ Trusting long-run volatility and half-life when $\alpha + \beta \approx 1$ without an interval
- ▣ Evaluating forecasts in sample, or with losses that are not robust to a noisy proxy
- ▣ Using an LR test for a parameter on the boundary (e.g. $\delta = 0$ with $\delta \geq 0$) with the usual χ^2 critical value, or reading a truncated bootstrap interval as a test
- ▣ Joining two markets on returns instead of prices: creates spurious multi-day returns

Take-Home Assignment (Team Project)

- Choose one series not covered in class (another index, an exchange rate, a commodity or a crypto-asset)
- Test for ARCH effects; fit GARCH, GJR and EGARCH with three innovation distributions; select by BIC and diagnostics
- Evaluate one-day and one-month forecasts out of sample against EWMA with QLIKE and Diebold–Mariano
- Report the half-life with a bootstrap interval; for an entropy-based alternative, see [Pele, Lazar & Dufour \(2017\)](#)
- Log every AI error you catch in `AI_ERRORS.md` (at least three per project)
- Deliver a Quantlet folder (code, Metainfo, charts) and three slides: evidence, interpretation, limitations

Further Study on Quantinar

Advanced courses on quantinar.com

- ☐  Statistics of Financial Markets
- ☐  Applied Time Series Analysis with Python

Code and data

- ☐ Seminar notebook (Colab); Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_05

References (1/3)

- ▣ Andrews, D.W.K. (2000). Inconsistency of the bootstrap when a parameter is on the boundary of the parameter space. *Econometrica*, 68(2), 399–405.
- ▣ Andrews, D.W.K. (2001). Testing when a parameter is on the boundary of the maintained hypothesis. *Econometrica*, 69(3), 683–734.
- ▣ Blasques, F., Francq, C., Laurent, S. (2023). Quasi score-driven models. *Journal of Econometrics*, 234(1), 251–275.
- ▣ Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3), 307–327.
- ▣ Bollerslev, T., Wooldridge, J.M. (1992). Quasi-maximum likelihood estimation and inference in dynamic models with time-varying covariances. *Econometric Reviews*, 11(2), 143–172.
- ▣ Diebold, F.X., Mariano, R.S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3), 253–263.
- ▣ Engle, R.F., Ng, V.K. (1993). Measuring and testing the impact of news on volatility. *Journal of Finance*, 48(5), 1749–1778.

References (2/3)

- ▣ Fan, J., Qi, L., Xiu, D. (2014). Quasi-maximum likelihood estimation of GARCH models with heavy-tailed likelihoods. *Journal of Business & Economic Statistics*, 32(2), 178–191.
- ▣ Franke, J., Härdle, W.K., Hafner, C.M. (2019). *Statistics of Financial Markets: An Introduction*, 5th ed. Springer.
- ▣ Giacomini, R., White, H. (2006). Tests of conditional predictive ability. *Econometrica*, 74(6), 1545–1578.
- ▣ Glosten, L.R., Jagannathan, R., Runkle, D.E. (1993). On the relation between the expected value and the volatility of the nominal excess return on stocks. *Journal of Finance*, 48(5), 1779–1801.
- ▣ Hall, P., Yao, Q. (2003). Inference in ARCH and GARCH models with heavy-tailed errors. *Econometrica*, 71(1), 285–317.
- ▣ Hansen, B.E. (1994). Autoregressive conditional density estimation. *International Economic Review*, 35(3), 705–730.
- ▣ Hansen, P.R., Lunde, A., Nason, J.M. (2011). The model confidence set. *Econometrica*, 79(2), 453–497.

References (3/3)

- ▣ J.P. Morgan/Reuters (1996). *RiskMetrics — Technical Document*, 4th ed. New York.
- ▣ Li, W.K., Mak, T.K. (1994). *On the squared residual autocorrelations in non-linear time series with conditional heteroskedasticity*. *Journal of Time Series Analysis*, 15(6), 627–636.
- ▣ Mincer, J., Zarnowitz, V. (1969). *The evaluation of economic forecasts*. In J. Mincer (ed.), *Economic Forecasts and Expectations: Analysis of Forecasting Behavior and Performance*, NBER.
- ▣ Patton, A.J. (2011). *Volatility forecast comparison using imperfect volatility proxies*. *Journal of Econometrics*, 160(1), 246–256.
- ▣ Pele, D.T., Lazar, E., Dufour, A. (2017). *Information entropy and measures of market risk*. *Entropy*, 19(5), 226.